

Machine Learning for Natural Language Understanding

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Sources

- Multilingual
 - Language-specific LLM for Low-Resource Languages
 - <https://pieter.ai/trans-tokenization/>
 - European Languages
 - <https://opengpt-x.de/en/models/teuken-7b/>
 - Massively Multilingual
 - <https://aclanthology.org/2025.coling-main.428/>
- Multimodal
 - <https://mistral.ai/news/pixtral-12b/>
- Multitool
 - https://reports.weforum.org/docs/WEF_AI_Agents_in_Action_Foundations_for_Evaluation_and_Governance_2025.pdf
 - <https://modelcontextprotocol.io/docs/getting-started/intro>
- Multiagent
 - <https://www.byhand.ai/p/llm-vs-rag-vs-agent-workbook>
 - <https://developers.googleblog.com/en/a2a-a-new-era-of-agent-interoperability/>

Flipped Classroom Questions

- Name technical challenges of multilingual LLMs
- What core component needs to be added to make an LLM multimodal?
- How does an MCP-Client find out if there is a suitable tool provided by an MCP-Server?
- What is the difference wrt inference of “LLMs”, “RAG”, “Tool Use” and “Multi agent” with respect to their control logic?

“The future of language is about the development of language models...”¹

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